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ORIGINAL ARTICLE

SfMDiffusion: self-supervised monocular depth estimation in endoscopy based on diffusion models

Yu Li[1] · Da Chang[2] · Die Luo[3] · Jin Huang[1] · Lan Dong[4] · Du Wang[1] · Liye Mei[1,2,3] · Cheng Lei[1,5,6]

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Abstract

Purpose In laparoscopic surgery, accurate 3D reconstruction from endoscopic video is crucial for effective image-guided techniques. Current methods for monocular depth estimation (MDE) face challenges in complex surgical scenes, including limited training data, specular reflections, and varying illumination conditions.

Methods We propose SfMDiffusion, a novel diffusion-based self-supervised framework for MDE. Our approach combines: (1) a denoising diffusion process guided by pseudo-ground-truth depth maps, (2) knowledge distillation from a pre-trained teacher model, and (3) discriminative priors to enhance estimation robustness. Our design enables accurate depth estimation without requiring ground-truth depth data during training.

Results Experiments on the SCARED and Hamlyn datasets demonstrate that SfMDiffusion achieves superior performance: an Absolute relative error (Abs Rel) of 0.049, a Squared relative error (Sq Rel) of 0.366, and a Root Mean Square Error (RMSE) of 4.305 on SCARED dataset, and Abs Rel of 0.067, Sq Rel of 0.800, and RMSE of 7.465 on Hamlyn dataset.

Conclusion SfMDiffusion provides an innovative approach for 3D reconstruction in image-guided surgical techniques. Future work will focus on computational optimization and validation across diverse surgical scenarios. Our code is available at https://github.com/Skylanding/SfM-Diffusion.

Keywords Monocular depth estimation · Self-supervised learning · Diffusion · Discriminative prior

Introduction

In laparoscopic surgery, image-guided techniques are becoming increasingly important, yet challenges with endoscopic views and localization persist [1, 2]. Computer vision-based augmented reality (AR) systems provide a reliable solution by aligning preoperative CT scan data with real-time endoscopic video [3]. The accuracy of these systems heavily relies on the 3D reconstruction of the endoscopic video. However, accurate depth estimation in endoscopic scenes faces several unique challenges. The wet tissue surfaces create non-Lambertian reflections and specular highlights that violate photometric consistency assumptions. Rapid camera motion and tissue deformation cause significant motion blur and scene changes between frames. Moreover, varying illumination conditions and shadows affect feature matching reliability, while surgical instruments frequently cause occlusions and scene discontinuities [4].

In recent years, deep learning methods have been widely applied to monocular depth estimation (MDE). These methods can be broadly categorized into supervised, semi-supervised, and self-supervised approaches based on their requirements for ground truth data. Fully supervised methods, like convolutional neural networks, have achieved reliable results in depth map prediction [5, 6]. These methods typically rely on end-to-end training with paired RGB-depth data, achieving high accuracy in well-controlled environ-

✉ Liye Mei
liyemei@whu.edu.cn

✉ Cheng Lei
leicheng@whu.edu.cn

1 The Institute of Technological Sciences, Wuhan University, Wuhan, China

2 Department of Networked Intelligence, Pengcheng Laboratory, Shenzhen, China

3 School of Computer Science, Hubei University of Technology, Wuhan, China

4 Department of Gynecology, Renmin Hospital of Wuhan University, Wuhan, China

5 Suzhou Institute of Wuhan University, Suzhou, China

6 Shenzhen Institute of Wuhan University, Shenzhen, China

ments. However, they face significant limitations in surgical scenarios due to their dependence on large-scale annotated datasets. Semi-supervised approaches attempt to balance this limitation by combining limited ground truth depth with auxiliary supervision signals [7, 8], but still require considerable manual annotation effort.

Given the scarcity of ground truth depth data in surgical settings, self-supervised learning approaches have emerged as promising alternatives. Many self-supervised monocular depth estimation networks have been proposed, such as SfMLearner [9], Monodepth2 [10], SC-SfMLearner [11], and Endo-SfM [12]. These self-supervised methods use parallax information between adjacent frames to train the neural network and can generate relative depth maps. Nevertheless, these methods still struggle with the unique challenges of surgical environments. Traditional photometric loss-based approaches are sensitive to illumination variations and specular reflections common in endoscopic videos, while their underlying assumption of scene rigidity is frequently violated by tissue deformations. Furthermore, they lack robust mechanisms to handle rapid view changes in surgical procedures.

The emergence of diffusion models has opened new possibilities for MDE. Recent works have demonstrated their effectiveness in natural scene depth estimation [13], leveraging their ability to model complex depth distributions. Diffusion models are particularly promising for surgical MDE due to several key advantages: their iterative refinement process naturally handles uncertainty in depth estimation caused by specular reflections and deformations, while their denoising mechanism provides inherent robustness to illumination variations. However, existing diffusion-based approaches primarily focus on fully supervised settings with natural scenes, leaving the specific challenges of surgical environments unaddressed. To address these challenges while leveraging the advantages of diffusion models in surgical MDE, we propose SfMDiffusion, which refines a random depth distribution step-by-step through multiple denoising stages guided by visual conditions. Our approach specifically targets surgical scene challenges through a combination of innovative components designed to handle specular reflections, scene deformations, and illumination variations. Since ground truth depth for endoscopy is not available during training, we draw on the experience of the MonoDiffusion [14] and introduce a pseudo-true depth diffusion process to assist the diffusion process [15]. Specifically, this process gradually adds noise to the depth generated by a pre-trained AF-SfMLearner teacher model [16], which we refer to as pseudo-ground truth. This approach allows us to impose a knowledge distillation loss on the denoised depth to improve results [17]. To enhance robustness against specular reflections and unreliable regions, we apply a multi-view check filter to remove incorrect depths. Additionally, we incorporate the depth from AF-SfMLearner as a discriminative prior [18] input into the diffusion model, helping maintain geometric consistency despite scene deformations and illumination changes.

To summarize, our contributions are as follows:

- We propose SfMDiffusion, the first self-supervised diffusion-based framework for monocular depth estimation in endoscopic scenes. Our novel integration of denoising diffusion models with self-supervised learning achieves robust performance in surgical environments without requiring ground truth depth data.
- We design a pseudo-ground-truth diffusion process that leverages a pre-trained teacher model, enabling effective training in the absence of real ground truth depth data through knowledge distillation.
- We introduce a discriminative prior mechanism guided by teacher model predictions and multi-view consistency check to effectively handle specular reflections and scene deformations in surgical videos.
- Through extensive experiments on the SCARED and Hamlyn datasets, we demonstrate that SfMDiffusion achieves significant improvements over state-of-the-art methods while maintaining computational efficiency.

Methodology

Preliminaries

Self-supervised depth estimation

MDE aims to infer depth from a single image. Traditional supervised depth estimation methods, while effective, rely heavily on the availability and quality of ground-truth depth data. Self-supervised depth estimation offers a compelling alternative that eliminates the need for expensive ground-truth data and instead uses stereo image pairs or consecutive frames from a video to generate a supervisory signal. Initial approaches train separate networks for depth and pose estimation, leveraging the predicted depth and 6-DoF pose for view synthesis [19]. Further advances and innovations have occurred through discrete disparity prediction, self-reference refinement, and combining CNNs and Transformers into efficient hybrid architectures [20].

In endoscopic scenarios, depth and motion estimation face unique challenges due to narrow field of view, complex lighting conditions, specular reflections, and highly similar tissue structures. Several key approaches have been proposed to address these challenges: Monodepth2 [10] enhances accuracy by combining disparity and depth maps, while SC-SfMLearner [11] employs a spatial transformer-based module to improve training stability and depth prediction.

Diffusion models

Diffusion models represent a class of latent variable models that have demonstrated remarkable success in generative modeling [21]. These models operate by learning to reverse a gradual noise corruption process applied to data. For surgical depth estimation, this iterative denoising approach offers an effective mechanism to address the inherent uncertainties in predicting tissue geometry and surface variations.

The diffusion process systematically transforms an initial data distribution x_0 into a sequence of increasingly noisy latent samples, culminating in x_τ. The forward process follows:

$$q(x_\tau|x_0) = \mathcal{N}(x_\tau; \sqrt{\alpha_\tau}x_0, (1-\alpha_\tau)I), \tag{1}$$

where the noise intensity is controlled by:

$$\alpha_\tau = \prod_{n=0}^{\tau}(1-\beta_n), \tag{2}$$

generating noisy samples via:

$$x_\tau = \sqrt{\alpha_\tau}x_0 + \sqrt{1-\alpha_\tau}\epsilon, \quad \epsilon \sim \mathcal{N}(0,1), \tag{3}$$

where τ indexes the T diffusion steps, β_n defines the variance schedule, and $\mathcal{N}$ denotes Gaussian noise. In surgical applications, careful scheduling of noise addition is crucial for handling regions of varying uncertainty, particularly in areas affected by tissue deformation or specular highlights.

To recover clean depth maps, we employ a noise predictor $\epsilon_\theta(x_\tau, \tau)$ that learns the reverse diffusion dynamics. The denoising transition follows a Gaussian model:

$$p_\theta(x_{\tau-1}|x_\tau) = \mathcal{N}(x_{\tau-1}; \mu_\theta(x_\tau, \tau), \sigma_\tau^2 I), \tag{4}$$

with mean $\mu_\theta(x_\tau, \tau)$ derived from the predicted noise and current state, and variance σ_τ^2 governing the transition uncertainty.

SfMDiffusion

Model architecture

As illustrated in Fig. 1, the SfMDiffusion model comprises two primary modules: the *Teacher* and the *Student*, where the teacher module's depth maps and multi-view filters guide the student module's denoising process through visualization conditions and knowledge distillation.

The *Teacher* module adopts AF-SfMLearner's architecture, which employs a multi-scale approach for full-resolution depth prediction, leveraging spatial transformer's gradient locality for improved training stability, while its Motion Module estimates rigid flow to account for camera motion, enabling consistent view synthesis. These capabilities, combined with its proven effectiveness in handling surgical scenes, make AF-SfMLearner an ideal choice for our teacher model. Building upon this foundation, we enhance the depth prediction reliability through multi-view consistency checking that validates depth estimates across adjacent frames. This is complemented by a confidence-based filtering mechanism that identifies potentially unreliable regions affected by specular reflections and tissue deformations, ensuring robust depth maps that serve as pseudo-ground-truth for the diffusion process and knowledge distillation.

The *Student* module contains two branches. The first branch features an encoder derived from the DepthNet encoder(Lite-Mono [22]), which extracts multi-scale features from the input image. The features are processed through 3×3 convolutional layers and upsampling layers to form visualization conditions that guide the diffusion denoising process. The second branch in the student module employs an encoder identical to the DepthNet encoder from the teacher module(Lite-Mono-8 M [22]), including pre-trained weights. The encoder generates visualization conditions used to enhance the diffusion model's decision-making priors. These priors are integrated through feature fusion layers where learnable attention weights α_t modulate their influence based on the current denoising step t and local image features. The student module includes a lightweight noise predictor consisting of an embedding layer and three 3×3 convolutional layers. At each denoising step, the predictor processes multi-scale feature tokens from both branches, with the attention weights α_t determining their relative contributions to the noise estimation.

For pose estimation, we adopt a network architecture similar to prior works [22], utilizing a pre-trained ResNet18 as the encoder followed by four convolutional layers as the decoder. It takes stacked pairs of images as input and estimates the 6-DOF relative pose between them.

Depth estimation for diffusion denoising

In the overall SfMDiffusion architecture, the task of MDE is reframed as an iterative denoising process. Given a target frame f_t, our goal is to predict a depth map D_t. This process takes place primarily in the *Student* module, which utilizes the multi-scale features extracted by the encoders to refine depth estimates under the guidance of the teacher's depth condition.

The denoising process follows a reverse diffusion process with T steps. At each step τ, the model refines the depth estimate according to:

$$p_\theta(D_{\tau-1}|D_\tau, c) = \mathcal{N}\left(D_{\tau-1}; \mu_\theta(D_\tau, \tau, c), \sigma_\tau^2\mathbf{I}\right), \tag{5}$$

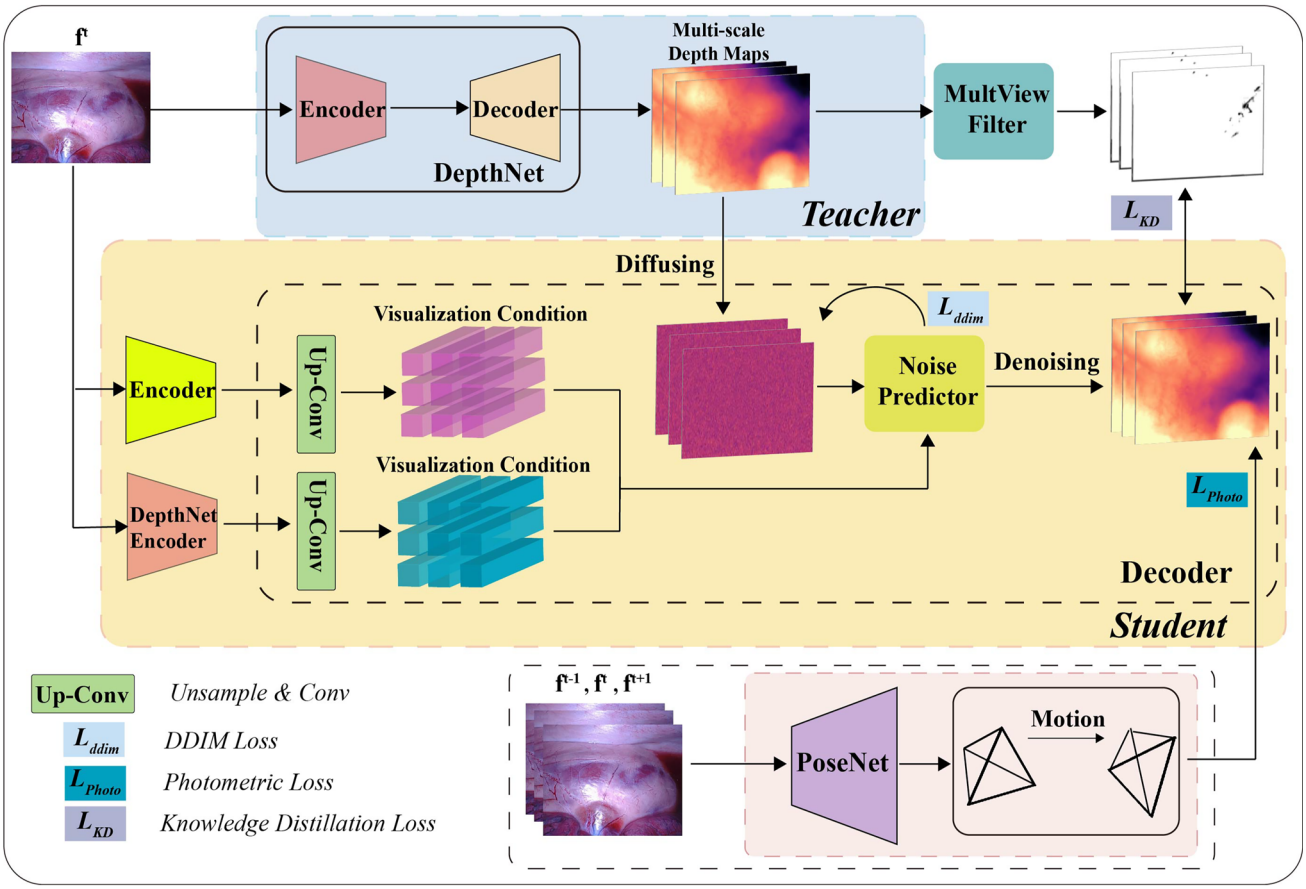

Fig. 1 An overview of our proposed SfMDiffusion. SfMDiffusion employs an additional teacher model to aid in propagation during the training phase. The teacher model is self-supervised and pre-trained using AF-SfMLearner. Upon completion of the training, only the DepthNet-related weights pre-trained by ResNet are retained

where c represents the conditions derived from the teacher model's depth predictions and multi-scale features. The mean μ_θ is predicted by our noise predictor network, which takes as input the noisy depth D_τ, the step embedding τ, and the condition c. The variance σ_τ^2 follows a fixed schedule to gradually reduce noise in the prediction.

Pseudo-ground-truth diffusion

In the absence of true ground-truth depth for self-supervised MDE, the *Teacher* module plays a crucial role by generating pseudo-ground-truth depth maps. We pre-train a teacher model based on the AF-SfMlearner framework, which provides these pseudo-ground-truth depth maps $D_{\text{pseudo}}(p)$. For each input frame, the teacher model generates multi-scale depth predictions that are validated through our multi-view consistency check. It compares the depth predictions across consecutive frames using the estimated camera poses, filtering out pixels where the reprojected depth values show high variance.

The diffusion process within the student module is defined as:

$$q(D_\tau | D_{\text{pseudo}}) = \mathcal{N}\left(D_\tau \mid \sqrt{\alpha_\tau} D_{\text{pseudo}}, (1-\alpha_\tau)\mathbf{I}\right). \quad (6)$$

The multi-view check filter $\Phi(p)$ measures the consistency of depth predictions across multiple views, where $\Phi(p) = 1$ when the standard deviation of reprojected depths at pixel p is below a threshold ϵ, and 0 otherwise. This filter is applied in the knowledge distillation loss:

$$L_{\text{KD}} = \sum_p \Phi(p) \odot (D_t(p) - D_{\text{pseudo}}(p)), \quad (7)$$

where regions with unreliable pseudo-ground-truth are automatically excluded from the supervision signal, particularly in areas with specular reflections or rapid tissue deformations, where depth predictions are less reliable.

Discriminative prior enhancement

To further enhance the denoising capability within the *Student* module, we leverage the depth predictions from the *Teacher* module as discriminative priors [23]. Different from the pseudo-ground-truth which provides direct supervision, these priors serve as dynamic guidance during the iterative denoising process. The prior condition, represented as D_{teacher} from the AF-SfMlearner model, modifies the conditional probability distribution of the denoising process:

$$p_\theta(D_{\tau-1}|D_\tau, D_{\text{teacher}}) = \mathcal{N}\left(D_{\tau-1}; \mu_\theta(D_\tau, \tau, D_{\text{teacher}}), \sigma_\tau^2 \mathbf{I}\right). \tag{8}$$

The mean function μ_θ determines how the prior information influences the denoising step. At each step τ, the model learns to adjust the noisy depth D_τ toward a more accurate estimate by considering both the current state and the structural information encoded in D_{teacher}. The conditional denoising process effectively incorporates the teacher's knowledge while maintaining the flexibility to deviate from it when necessary, as determined by the learned parameters θ.

Overall loss function

Our SfMDiffusion model incorporates multiple supervision signals during training: photometric consistency between frames, knowledge distillation from the teacher's pseudo-ground-truth depth maps, and diffusion trajectory guidance enhanced by discriminative priors. The total loss function combines three components: the photometric loss, the knowledge distillation loss, and the DDIM loss.

The photometric loss L_{ph} measures the consistency between a synthesized target frame $f_{s\to t}$ and the actual target frame f_t, where $f_{s\to t}$ is generated by warping the source frame f_s using the estimated depth map D_t. The loss combines structural similarity and L1 distance:

$$L_{\text{ph}} = \sum_p \left[\kappa \frac{1 - \text{SSIM}(f_t(p), f_{s\to t}(p))}{2} + (1-\kappa)\|f_t(p) - f_{s\to t}(p)\|_1\right], \tag{9}$$

where SSIM is the Structural Similarity Index, and $f_{s\to t}$ represents the synthesized target frame using the source frame f_s and the estimated depth map D_t. The SSIM term helps preserve structural information and edge alignment in the depth prediction, while the L1 term enhances depth accuracy in textureless regions. The combination of both terms enables robust depth estimation across various surgical scene textures.

The knowledge distillation loss L_{KD} is applied to ensure that the depth maps predicted by the student module align with the pseudo-ground-truth depth maps D_{pseudo} provided by the teacher module. This loss enables the student model to learn from the teacher model's refined depth estimates, enhancing its own prediction accuracy. The knowledge distillation loss is defined as:

$$L_{\text{KD}} = \sum_p \Phi(p) \odot (D_t(p) - D_{\text{pseudo}}(p)), \tag{10}$$

where $\Phi(p)$ is the multi-view check filter that mitigates errors in the pseudo-ground-truth depth maps. This loss function ensures that the student model benefits from the teacher model's pre-trained knowledge, leading to improved depth prediction performance. By selectively applying supervision through the multi-view check filter, the loss effectively guides the student model while avoiding error propagation from regions with uncertain depth estimates.

The DDIM loss L_{DDIM} is designed to ensure that the denoising process within the student module aligns with the diffusion model. It encourages the predicted depth at each diffusion step τ to follow the expected diffusion trajectory. The DDIM loss is defined as:

$$L_{\text{DDIM}} = \sum_\tau \|D_\tau - \hat{D}_\tau\|^2, \tag{11}$$

where $\hat{D}_\tau$ is the predicted depth at step τ. The DDIm loss is essential for maintaining consistency in the iterative denoising process, ensuring that each step brings the predicted depth closer to the final accurate estimate. The quadratic form of the loss places stronger penalties on large deviations from the expected trajectory, promoting stable and gradual refinement of depth predictions throughout the denoising process.

Overall loss L_{total} is then formulated as a weighted combination of the photometric loss, knowledge distillation loss, and DDIM loss.

$$L_{\text{total}} = \lambda_1 L_{\text{ph}} + \lambda_2 L_{\text{KD}} + \lambda_3 L_{\text{DDIM}}, \tag{12}$$

where λ_1, λ_2, and λ_3 are all set to 1 through experimental validation. Following previous works [10], we utilize dual source frames when computing the photometric loss L_{ph}, selecting the frame that yields the lower loss value to effectively handle cases where occlusion occurs.

Experiments and results

Datasets and evaluation metrics

Extensive experiments are conducted on two surgical datasets:

- SCARED [24]. The SCARED dataset comprises 35 endoscopic videos of fresh porcine cadaver abdominal anatomy collected using a Da Vinci Xi endoscope, providing point cloud and ego-motion ground truth. Following a split protocol inspired by the Eigen-Zhou evaluation protocol from the KITTI benchmark [25], the dataset is divided into 15,351 frames for training, 1,705 frames for validation, and 551 frames for testing.
- Hamlyn.[1] The Hamlyn dataset contains in vivo endoscopic videos recorded during minimally invasive surgeries using da Vinci robot, featuring various surgical procedures with point cloud ground truth. Following the same protocol as SCARED, the dataset is divided into 12,853 frames for training, 1428 frames for validation, and 476 frames for testing.

The evaluation utilizes multiple metrics consistent with prior depth estimation works:

- Abs Rel: $\frac{1}{|D|}\sum_{d\in D}\frac{|d_t-d|}{d}$;
- Sq Rel: $\frac{1}{|D|}\sum_{d\in D}\frac{||d_t-d||^2}{d}$;
- RMSE: $\sqrt{\frac{1}{|D|}\sum_{d\in D}||d_t-d||^2}$;
- RMSE log: $\sqrt{\frac{1}{|D|}\sum_{d\in D}||\log d_t-\log d||^2}$;
- $\delta <$ thr: % of d satisfies $(\max(\frac{d_t}{d},\frac{d}{d_t})=\delta <$ thr) for thr $= 1.25, 1.25^2, 1.25^3$.

Among these metrics, Abs Rel, Sq Rel, RMSE and RMSE log are depth error metrics where lower values indicate better performance. The $\delta < 1.25$, $\delta < 1.25^2$ and $\delta < 1.25^3$ are depth accuracy metrics where higher values indicate better performance. Due to the inherent scale ambiguity in self-supervised MDE systems, the median scaling technique is employed to recover the absolute scale for depth evaluation [26].

Implementation details

Our method is implemented on the PyTorch platform (version 2.0.1) and trained on an NVIDIA GeForce RTX 3090 GPU. The system runs on Ubuntu 20.04 with CUDA 11.8. We use the AdamW optimizer with a weight decay of 1e-2 and the batch size is set to 15. When training the model from scratch, an initial learning rate of 1e-4 is adopted along with a cosine learning rate schedule. We explored different numbers of diffusion steps during training (500, 1000, 2000, and 3000 steps) and found that 2000 steps achieved the best trade-off between image quality and computational cost. While fewer steps reduced training time, they led to notable degradation in generated image quality. During inference, we adopt an improved sampling procedure that enables using only 20 denoising steps while maintaining generation quality comparable to standard sampling methods.

To assess the practical applicability of our approach, we analyzed the model size and computational efficiency. Our SfMDiffusion model contains approximately 12.3M parameters, which is comparable to AF-SfMLearner's 11.68M parameters. In terms of inference speed, our model achieves an average inference time of 11.2ms per frame on an NVIDIA RTX 3090 GPU, slightly higher than AF-SfMLearner's 9.9ms, within the requirements for real-time applications.

Results

Quantitative results

Our proposed method, SfMDiffusion, demonstrates superior performance when compared to several state-of-the-art depth estimation methods on both the SCARED and Hamlyn datasets. As shown in Table 1, on the SCARED dataset, SfMDiffusion achieves the best results in most metrics, with the lowest Abs Rel and Sq Rel scores, indicating minimal absolute and relative errors. While Surgical-DINO achieves slightly better RMSE and RMSE log scores, our method remains highly competitive in these metrics. For the accuracy metric $\delta < 1.25$, SfMDiffusion ties with Surgical-DINO for the highest score, outperforming other methods, with most predicted depths falling within a factor of 1.25 of the ground truth depths. Furthermore, Table 2 shows that SfMDiffusion achieves consistent superior performance on the Hamlyn dataset, outperforming all compared methods across all metrics, including Abs Rel, Sq Rel, RMSE, RMSE log and accuracy score $\delta < 1.25$. These results demonstrate the robustness and generalization capability of our proposed method across different surgical scenarios.

Qualitative results

We also present qualitative comparisons of depth estimation results in Fig. 2. As shown in the visualization, SfMDiffusion demonstrates superior performance in capturing the intricate details of endoscopic scenes. Compared to previous methods, our approach more accurately preserves the geometric structure of anatomical features, particularly in challenging areas with complex tissue textures and varying illumination.

Ablation study

To better understand the impact of each component in the SfMDiffusion framework, we conducted comprehensive ablation studies (Table 3). To demonstrate the generalization capability of our method, we evaluate the performance on both SCARED and Hamlyn datasets.

[1] http://hamlyn.doc.ic.ac.uk/vision/.

Table 1 Quantitative depth comparison on the SCARED dataset

Method	Abs Rel ↓	Sq Rel ↓	RMSE ↓	RMSE log ↓	δ ↑
SFMlearner [9]	0.079	0.879	6.896	0.110	0.947
Fang [27]	0.078	0.794	6.794	0.109	0.946
DeFeat-Net [28]	0.077	0.792	6.688	0.108	0.941
SC-SfMLearner [11]	0.068	0.645	5.988	0.097	0.957
Monodepth2 [10]	0.071	0.590	5.606	0.094	0.953
Endo-SfM [12]	0.062	0.606	5.726	0.093	0.957
AF-SfMlearner [16]	0.059	0.435	4.925	0.082	0.974
DINOv2 [29]	0.060	0.459	4.692	0.081	0.963
Surgical-DINO [30]	0.053	0.377	**4.296**	**0.074**	**0.975**
SfMDiffusion(Ours)	**0.049**	**0.366**	4.305	0.078	**0.975**

The best results are in bold. The second-best results are in underlined

Table 2 Quantitative depth comparison on the Hamlyn dataset

Method	Abs Rel ↓	Sq Rel ↓	RMSE ↓	RMSE log ↓	δ ↑
SFMlearner [9]	0.090	1.359	10.876	0.147	0.910
Fang [27]	0.142	2.090	12.666	0.189	0.810
DeFeat-Net [28]	0.092	1.174	9.902	0.138	0.922
SC-SfMLearner [11]	0.081	0.975	9.130	0.121	0.944
Monodepth2 [10]	0.100	1.106	8.786	0.126	0.927
Endo-SfM [12]	0.073	0.977	8.960	0.111	0.962
AF-SfMlearner [16]	0.079	0.800	8.154	0.109	0.959
DINOv2 [29]	0.082	0.832	8.161	0.109	0.958
Surgical-DINO [30]	0.078	0.845	8.132	0.109	0.959
SfMDiffusion(Ours)	**0.067**	**0.800**	**7.465**	**0.106**	**0.962**

The best results are in bold. The second-best results are underlined

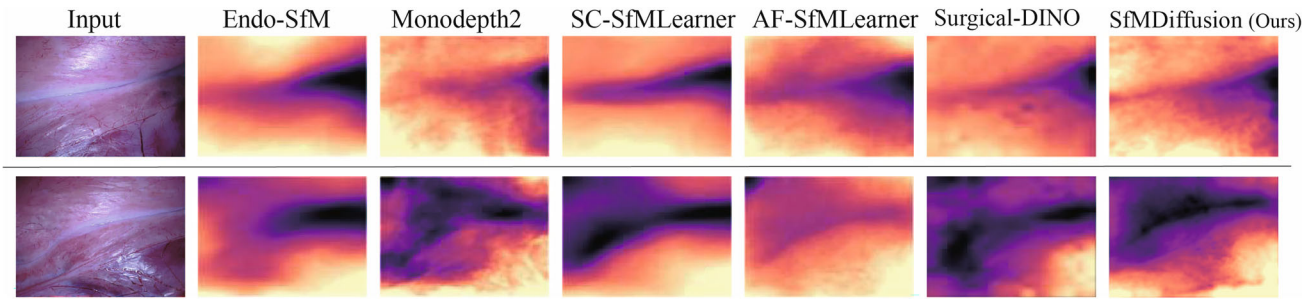

Fig. 2 Qualitative depth comparison on the SCARED dataset

Table 3 Ablation study on SCARED/Hamlyn datasets

ID	PGD	DPE	L_{KD}	L_{ph}	Abs Rel ↓	RMSE ↓	δ ↑
1	✓				0.062/0.079	5.270/8.154	0.963/0.955
2	✓	✓			0.058/0.074	4.939/7.892	0.966/0.957
3	✓	✓	✓		0.053/0.072	4.443/7.823	0.972/0.958
4	✓	✓		✓	0.053/0.069	4.584/7.688	0.970/0.959
5	✓	✓	✓	✓	**0.049/0.067**	**4.305/7.465**	**0.975/0.962**

PGD pseudo-ground-truth diffusion, *DPE* discriminative prior enhancement; L_{KD}: The knowledge distillation loss; L_{ph}: The photometric loss
Bold represents the best results

Starting with the baseline model that utilizes only PGD (ID 1), we analyze the effectiveness of each proposed component. The addition of DPE (ID 2) brings consistent improvements on both datasets, demonstrating DPE's ability to capture discriminative features for distinguishing subtle depth variations in endoscopic scenes, particularly in regions with specular reflections and ambiguous textures. The knowledge distillation loss (ID 3) and photometric loss (ID 4) contribute complementary supervision signals to the framework. L_{KD} leverages teacher model knowledge to provide reliable depth cues in challenging cases where direct depth estimation becomes ambiguous. In parallel, L_{ph} enforces geometric consistency across different views, preserving fine-grained structural details. The distinct yet complementary roles of these two losses are evidenced by performance variations across different datasets.

The fully integrated model (ID 5), which combines all components, achieves superior performance on all metrics on both datasets, demonstrating that the proposed components are able to work together to address various aspects of the endoscopic depth estimation challenge. The consistent performance gains across different datasets further demonstrate the robustness and generalization capability of the SfMDiffusion framework .

Conclusion

In this study, we presented SfMDiffusion, a novel framework for self-supervised monocular depth estimation tailored for laparoscopic surgery. Our approach effectively addresses the challenges of depth estimation in endoscopic videos by utilizing a denoising diffusion process guided by pseudo-ground-truth data generated from a pre-trained AF-SfMLearner model. The integration of knowledge distillation and photometric loss further enhances the accuracy of our depth predictions. Experimental results on the SCARED and Hamlyn datasets confirm that SfMDiffusion outperforms existing methods, demonstrating its potential for improving 3D reconstruction in image-guided surgical applications.

While our approach shows promising results, we acknowledge several limitations. The reliance on pseudo-ground-truth data may introduce inherent biases from the teacher model, potentially affecting performance in scenarios that differ significantly from the training data. Additionally, the performance in real surgical environments with diverse lighting conditions and tissue deformations requires further validation.

Future work will focus on addressing these limitations through developing robust methods for handling tissue deformation and dynamic scene changes in surgical procedures. Additionally, exploring advanced self-supervised learning strategies to reduce the dependence on pseudo-ground-truth data would be valuable. These improvements would enhance the practical applicability of our framework in real surgical scenarios.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethics approval and consent to participate This article does not include any studies involving human participants or animals conducted by the authors. Additionally, it does not contain any patient data.

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